Cover Letter

November 6, 2026

Faculty Search Committee
David L. Payne Department of Electrical Engineering and Computer Science
University of Missouri
Columbia, Missouri

Re: Assistant Professor in Artificial Intelligence, Geospatial and National Security Applications (Job ID 61589)

Dear Members of the Search Committee:

I am applying for the Assistant Professor in Artificial Intelligence position in the David L. Payne Department of Electrical Engineering and Computer Science at the University of Missouri. I earned my PhD in Electrical and Computer Engineering from the University of Oklahoma in 2025, with a dissertation focused on reliable AI/ML for wireless networks. My research and industrial experience encompass data-efficient learning, generative and transfer learning, anomaly detection, reinforcement learning, intelligent sensing, and dependable AI systems. I am especially interested in bringing these methodological strengths to geospatial intelligence, multimodal sensing, and high-consequence decision support in collaboration with Missouri's Center for Geospatial Intelligence (CGI) and AI Education, Research and Infrastructure Center (AERI).

My doctoral research addressed a common challenge in applied AI: models that perform well in controlled experiments often degrade when data are scarce or operating conditions change. I developed domain-informed generative augmentation and transfer-learning approaches for propagation-model prediction, AI-driven outage detection and network self-healing, and reproducible wireless simulation frameworks. This work resulted in peer-reviewed publications in Computer Communications, Sensors, IEEE Network, and IEEE conferences, including PIMRC and VTC. Related projects in wearable sensing and time-series deep learning further strengthened my experience with noisy sensor streams, inter-user variability, and cross-domain evaluation.

In industry, I have worked with AT&T, AT&T Labs, Futurewei, and Impac Exploration Services on AI-enabled diagnostics, network measurements, and data engineering. At AT&T, I developed and evaluated machine-learning diagnostics for operational service assurance, improving automated root-cause classification accuracy from approximately 20% to 80%. At AT&T Labs, I analyzed more than 500,000 LTE/5G diagnostic logs to identify anomalies and support engineering decisions. These experiences have reinforced my focus on transparent evaluation, operational reliability, and translating research methods into useful systems.

My proposed research vision, described in the accompanying narrative, addresses reliable, data-efficient geospatial AI through three connected directions: robust visual feature detection under geographic and sensor shift; principled fusion of optical, radar, geolocation, and time-series information; and uncertainty-aware, resource-efficient AI for analyst-supported decisions. This is a deliberate extension of my prior AI research into geospatial applications, not a claim that I have previously led satellite-imagery or defense-funded projects. I would establish rigorous public-data benchmarks, develop collaborations with CGI and AERI researchers, and pursue eligible external funding opportunities as the program matures.

Previously, as a Lecturer at the University of Lahore, I taught Computer Programming, Communication Systems, and Wireless Communication; I also supported laboratory instruction at UET Taxila and mentored undergraduate and graduate research students at COMSATS. At Missouri, I would contribute to undergraduate and graduate teaching in artificial intelligence, machine learning, programming, data analytics, intelligent sensing, and networked systems, while supporting the new MS and PhD programs in AI through project-based mentoring and reproducible research.

I welcome the opportunity to contribute to Missouri's growing AI research community and interdisciplinary partnerships. Thank you for considering my application.

Sincerely,

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